

UC SAN DIEGO / ECE 251C

2

Continuous-domain signal processing

Dirac impulses, convolution, Fourier analysis, and differentiation

What changes when the signal's domain is continuous?

A signal's *domain* is the set on which it is defined. It need not represent time: a spatial profile depends on position along a line, and an image can be modeled on a plane. We therefore speak of *continuous-domain signal processing*. Here we work on $\mathbb{R}$ and write t for the coordinate, whether or not it represents time. Linearity, shifts, and impulse responses remain central. A moving average will connect the Dirac impulse to convolution and Fourier analysis.

2.1 Signals and systems on the real line

A discrete-domain signal has values $x[n]$ for $n \in \mathbb{Z}$. Here a continuous-domain signal has values $f(t)$ for $t \in \mathbb{R}$. The word *continuous* describes the domain, not the function: f may still have jumps. A system H maps an input f to an output $g = Hf$.

Linearity. If $g_1 = Hf_1$ and $g_2 = Hf_2$, then

$$H(\alpha f_1 + \beta f_2) = \alpha g_1 + \beta g_2 \quad (\alpha, \beta \in \mathbb{R}). \quad (2.1)$$

Translation invariance. Define $(T_a f)(t) := f(t - a)$. When t represents time, a positive a is a delay and we call this *time invariance*. In either interpretation,

$$H(T_a f) = T_a(Hf) \quad \text{for every } a \in \mathbb{R}. \quad (2.2)$$

Unlike the discrete case, the shift a may be *any real number*.

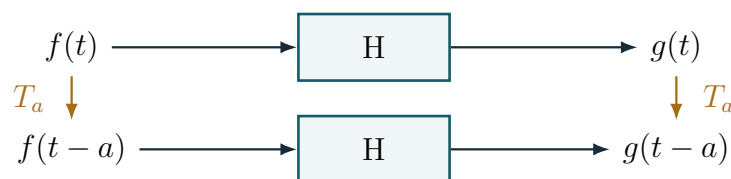


Figure 2.1: Translation invariance: shifting the input shifts the output by the same amount. For a temporal signal, the shift is a time shift.

2.2 The Dirac impulse: unit mass at one point

In discrete time, the Kronecker impulse satisfies $\delta[0] = 1$ and $\delta[n] = 0$ for $n \neq 0$. Copying this definition to real time does *not* give a unit-area impulse: an ordinary function supported at one point has integral zero.

Instead, we define the *Dirac impulse* by what it does when integrated against a signal. For a function φ that is continuous at a ,

$$\int_{\mathbb{R}} \varphi(t) \delta(t - a) dt = \varphi(a). \quad (2.3)$$

This is the *sifting property*. The impulse at a evaluates a function at a . The integral is notation for this action, not an ordinary integral against an infinitely tall function. We can think of the Dirac impulse as a unit point mass: all its weight is concentrated at one location.

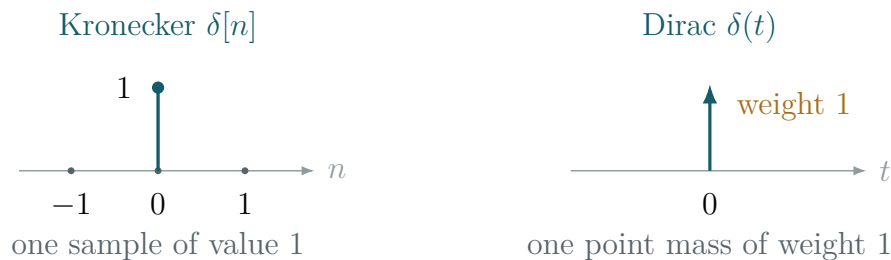


Figure 2.2: The arrow for a Dirac impulse records its location and weight. Its drawn height is not the value of $\delta(0)$.

A limiting picture. For $\varepsilon > 0$, let $r_\varepsilon(t) = \varepsilon^{-1}$ on $[-\varepsilon/2, \varepsilon/2]$ and zero elsewhere. Its width is ε , its height is $1/\varepsilon$, and its area is one. Then

$$\begin{aligned} \left| \int_{\mathbb{R}} r_\varepsilon(t) \varphi(t) dt - \varphi(0) \right| &\leq \frac{1}{\varepsilon} \int_{-\varepsilon/2}^{\varepsilon/2} |\varphi(t) - \varphi(0)| dt \\ &\leq \sup_{|t| \leq \varepsilon/2} |\varphi(t) - \varphi(0)| \rightarrow 0. \end{aligned}$$

Continuity at zero proves the limit. The integrals approach $\varphi(0)$ for every function φ continuous at zero. This is how the pulses approach δ ; they do not converge pointwise to an ordinary unit-area function.

2.3 Three consequences of sifting

Each identity below follows from the defining action; none requires assigning a pointwise value to δ .

1. The impulse has total mass one. Use the constant function $\varphi(t) = 1$:

$$\int_{\mathbb{R}} \delta(t) dt = \int_{\mathbb{R}} 1 \cdot \delta(t) dt = 1. \quad (2.4)$$

More generally, $c\delta(t - a)$ has weight c and evaluates φ as $c\varphi(a)$. A negative weight is allowed; it means subtraction, not a negative time location.

2. Its Fourier transform is constant. We use angular frequency ω and the convention

$$\hat{f}(\omega) := \int_{\mathbb{R}} f(t) e^{-j\omega t} dt, \quad j^2 = -1. \quad (2.5)$$

For an ordinary signal, absolute integrability is a sufficient condition for this integral. For the impulse, the integral is evaluated by sifting:

$$\hat{\delta}(\omega) = \int_{\mathbb{R}} \delta(t) e^{-j\omega t} dt = e^{-j\omega \cdot 0} = 1. \quad (2.6)$$

The same calculation at a gives $\widehat{\delta(\cdot - a)}(\omega) = e^{-j\omega a}$. A delay changes the phase, but not the magnitude, of this spectrum.

3. Convolution with an impulse leaves the signal unchanged. Continuous-time convolution is

$$(h * f)(t) := \int_{\mathbb{R}} h(u) f(t - u) du$$

whenever the integral is defined. For an impulse kernel, apply sifting to the function $u \mapsto f(t - u)$:

$$(\delta * f)(t) = \int_{\mathbb{R}} \delta(u) f(t - u) du = f(t). \quad (2.7)$$

Likewise, a shifted impulse gives $(\delta(\cdot - a) * f)(t) = f(t - a)$. Thus δ is the impulse response of the identity system, and $\delta(\cdot - a)$ is the impulse response of a pure delay.

One definition, three interpretations

The Dirac impulse has unit mass, a flat Fourier spectrum, and the identity action under convolution. These are different consequences of the same sifting rule.

2.4 From impulse response to convolution

The *impulse response* of a continuous-time system is $h := H\delta$. Time invariance specifies the response to an impulse at any other location:

$$H\{\delta(\cdot - u)\}(t) = h(t - u).$$

Linearity then gives the response to a finite collection of weighted impulses. For example,

$$H\{2\delta + \delta(\cdot - a)\}(t) = 2h(t) + h(t - a).$$

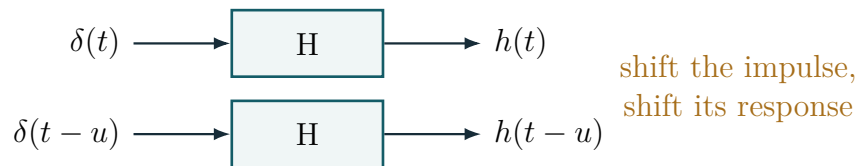


Figure 2.3: Every translated impulse produces a translated copy of the same response.

Sifting also writes a continuous input as an integral superposition:

$$f(t) = \int_{\mathbb{R}} f(u) \delta(t - u) du. \quad (2.8)$$

Assuming the system respects this integral superposition, linearity and time invariance give

$$\begin{aligned} (Hf)(t) &= \int_{\mathbb{R}} f(u) H\{\delta(\cdot - u)\}(t) du \\ &= \int_{\mathbb{R}} f(u) h(t - u) du \\ &= \int_{\mathbb{R}} h(v) f(t - v) dv = (h * f)(t), \end{aligned} \quad (2.9)$$

where the last line uses $v = t - u$. Convolution is a weighted sum of shifted impulse responses, now with an integral in place of the discrete sum.

What justifies the integral step? Algebraic linearity concerns finite sums. The impulse-response derivation also assumes that the system respects the limit used to form the integral superposition. Alternatively, the convolution formula directly defines a linear system. The identity

$$(h * T_a f)(t) = \int_{\mathbb{R}} h(v) f(t - a - v) dv = (h * f)(t - a)$$

proves time invariance. If $|f(t)| \leq M$ and $\int_{\mathbb{R}} |h(v)| dv < \infty$, then

$$\int_{\mathbb{R}} |h(v) f(t - v)| dv \leq M \int_{\mathbb{R}} |h(v)| dv < \infty.$$

This is a sufficient condition for the convolution to converge absolutely.

2.5 Support, closure, and compactness

The ordinary function $h(t)$ is also called the *filter*. Its *support* is

$$\text{supp}(h) := \overline{\{t \in \mathbb{R} : h(t) \neq 0\}}. \quad (2.10)$$

The bar denotes *closure*. Intuitively, keep the set and add any missing points that can be approached arbitrarily closely from within it. For an open interval, this adds its two endpoints, but no points beyond them.

A set $S \subseteq \mathbb{R}$ is *closed* if it contains every limit of a convergent sequence of points in S . Its closure $\overline{S}$ includes S and all such limit points. Equivalently, $t \in \overline{S}$ if every interval $(t - \varepsilon, t + \varepsilon)$, however small, meets S .

For example, $a + (b - a)/m \in (a, b)$ for $m \geq 2$ and tends to a . Thus a belongs to the closure of (a, b) even though a does not belong to the interval itself. The same applies to b . Points outside $[a, b]$ have a small neighborhood that misses (a, b) , so

$$\overline{(a, b)} = [a, b].$$



Figure 2.4: Closure adds the missing limit points. Filled endpoints are included; open endpoints are excluded.

Exercise, with answers. Assume $a < b < c < d$.

Set	Closed?	Closure
(a, b)	No	$[a, b]$
$[a, b)$ or $(a, b]$	No	$[a, b]$
$[a, b]$	Yes	$[a, b]$
$(a, b) \cup [c, d)$	No	$[a, b] \cup [c, d]$

A set is *bounded* if it lies in some finite interval $[-B, B]$. In $\mathbb{R}$, a set is *compact* precisely when it is closed and bounded. This is the real-line Heine–Borel characterization; it is the characterization we use here. The support in (2.10) is already closed, so it is compact exactly when it is bounded.

2.6 Finite and infinite impulse responses

A continuous-time filter is *FIR* if $\text{supp}(h)$ is compact, and *IIR* if $\text{supp}(h)$ is unbounded. Here “finite” means *finite duration*, not a finite number of real-valued time instants.

We use the indicator notation

$$\mathbf{1}_A(t) := \begin{cases} 1, & t \in A, \\ 0, & t \notin A. \end{cases}$$

Consider the two filters

$$h_{\text{box}}(t) = \mathbf{1}_{[-1/2, 1/2]}(t), \quad h_{\text{tail}}(t) = e^{-t} \mathbf{1}_{[0, \infty)}(t).$$

Their supports are $[-1/2, 1/2]$ and $[0, \infty)$, respectively. The first filter is FIR. The second is IIR: its positive tail becomes small but never becomes identically zero.



Figure 2.5: A response can decay rapidly and still have infinite duration. FIR/IIR describes where the response is nonzero, not how large it is.

How does this recover the discrete definition? For a sequence, the support is $S = \{n \in \mathbb{Z} : h[n] \neq 0\}$. Any bounded subset of $\mathbb{Z}$ is finite: if $|n| \leq B$, only finitely many integers are possible. Conversely, every finite set is bounded. Moreover, every subset of $\mathbb{Z}$ is closed in $\mathbb{R}$: a convergent sequence of integers is eventually constant, since distinct integers are at least one unit apart. Consequently,

$$S \text{ compact} \iff S \text{ finite}.$$

Thus the continuous-time definition has exactly the same support-based interpretation as the FIR definition for discrete filters.

Duration is not causality

The centered box uses both past and future input values. It is FIR, but not causal as a real-time filter. Delaying the box by $1/2$ gives support $[0, 1]$ and a causal moving average. Compact support alone does not decide causality.

2.7 What does the box filter do?

Let $h(t) = \mathbf{1}_{[-1/2, 1/2]}(t)$. Substitution into convolution gives

$$\begin{aligned} g(t) &= (h * f)(t) = \int_{-1/2}^{1/2} f(t - u) \, du \\ &= \int_{t-1/2}^{t+1/2} f(v) \, dv. \end{aligned} \quad (2.11)$$

For the second equality, set $v = t - u$, so $dv = -du$ and the integration limits reverse. The interval has length one, so this integral is exactly the average value of f in a unit-width window centered at t .

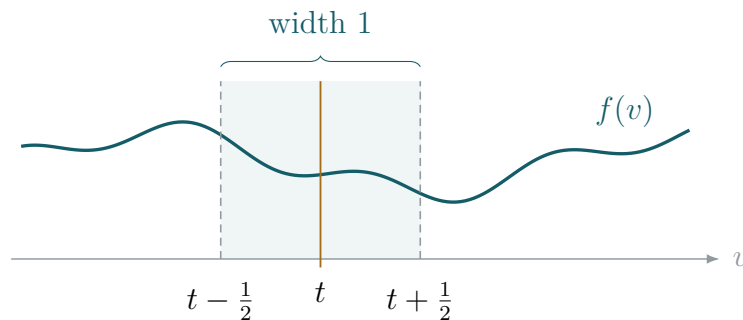


Figure 2.6: Move the window, integrate the enclosed signal, and record the result as $g(t)$. The filter is fixed; the window moves with the output time.

Worked example: a jump becomes a ramp. Let $f(t) = \mathbf{1}_{[0, \infty)}(t)$ be a unit step. The integral in (2.11) is the length of the part of $[t - 1/2, t + 1/2]$ lying to the right of zero. Therefore

$$g(t) = \begin{cases} 0, & t \leq -\frac{1}{2}, \\ t + \frac{1}{2}, & -\frac{1}{2} < t < \frac{1}{2}, \\ 1, & t \geq \frac{1}{2}. \end{cases} \quad (2.12)$$

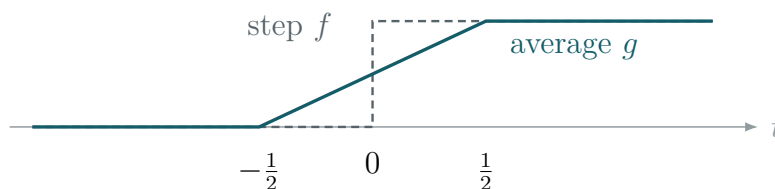


Figure 2.7: The dashed step is replaced by a linear transition of width one. Averaging preserves a constant region but spreads an abrupt change.

This explains the smoothing in time. To identify which oscillations are suppressed, we now compute the frequency response.

2.8 Pure frequencies are eigenfunctions

Extend a real linear filter to complex signals by acting separately on the real and imaginary parts. Euler's formula $e^{j\theta} = \cos \theta + j \sin \theta$ then lets us study both sinusoids at once.

Suppose $\int_{\mathbb{R}} |h(u)| du < \infty$. For the input $f(t) = e^{j\omega t}$,

$$\begin{aligned} (Hf)(t) &= \int_{\mathbb{R}} h(u) e^{j\omega(t-u)} du \\ &= e^{j\omega t} \int_{\mathbb{R}} h(u) e^{-j\omega u} du \\ &= \hat{h}(\omega) e^{j\omega t}. \end{aligned} \quad (2.13)$$

The integral is absolutely convergent because the exponential has magnitude one. Notice that the input itself need not have an absolutely convergent Fourier integral for this *filtering* calculation to be valid.

The function $\hat{h}(\omega)$ is the system's *frequency response*. In engineering notation, it is also written $H(j\omega)$:

$$H(j\omega) = \hat{h}(\omega) = \int_{\mathbb{R}} h(u) e^{-j\omega u} du. \quad (2.14)$$

Here H denotes the *system*, whereas $H(j\omega)$ denotes a *complex number* for each frequency.

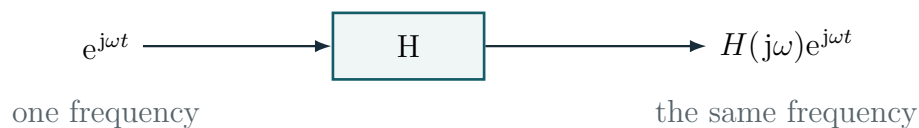


Figure 2.8: The exponential keeps its shape under an LTI filter. Its multiplier is an eigenvalue of the system.

If $H(j\omega) = \rho e^{j\theta}$ with $\rho \geq 0$, the output is $\rho e^{j(\omega t + \theta)}$. For a real filter, taking real parts yields

$$\cos(\omega t) \mapsto \rho \cos(\omega t + \theta).$$

Thus $|H(j\omega)|$ describes the amplitude gain; an argument of $H(j\omega)$ describes the phase shift when the response is nonzero.

Unlike discrete-time exponentials, continuous-time exponentials at ω and $\omega + 2\pi$ are generally different: $e^{j2\pi t}$ need not equal one for real t . The continuous-time frequency response need not be periodic.

2.9 Convolution becomes multiplication

The single-frequency calculation suggests that Fourier analysis should turn a convolution system into a collection of scalar multiplications. We can establish that directly, without assuming a Fourier inversion formula.

Proposition 2.1 (Convolution theorem). Suppose that h and f are absolutely integrable:

$$\int_{\mathbb{R}} |h(t)| dt < \infty, \quad \int_{\mathbb{R}} |f(t)| dt < \infty.$$

Then $g = h * f$ is defined almost everywhere (except on a set of length zero), is absolutely integrable, and

$$\hat{g}(\omega) = \hat{h}(\omega) \hat{f}(\omega). \quad (2.15)$$

Equivalently, $G(j\omega) = H(j\omega)F(j\omega)$.

Proof. First integrate the nonnegative absolute value; nonnegative integrands permit either order of integration:

$$\begin{aligned} \int_{\mathbb{R}} \int_{\mathbb{R}} |h(u)f(t-u)| du dt &= \int_{\mathbb{R}} |h(u)| \left(\int_{\mathbb{R}} |f(t-u)| dt \right) du \\ &= \left(\int_{\mathbb{R}} |h(u)| du \right) \left(\int_{\mathbb{R}} |f(v)| dv \right) < \infty. \end{aligned}$$

The substitution in the second line is $v = t - u$. The same bound controls $\int_{\mathbb{R}} |g(t)| dt$. Absolute integrability permits reversing the order of integration in the Fourier calculation:

$$\begin{aligned} \hat{g}(\omega) &= \int_{\mathbb{R}} \int_{\mathbb{R}} h(u)f(t-u)e^{-j\omega t} du dt \\ &= \int_{\mathbb{R}} h(u) \left(\int_{\mathbb{R}} f(v)e^{-j\omega(v+u)} dv \right) du \\ &= \left(\int_{\mathbb{R}} h(u)e^{-j\omega u} du \right) \left(\int_{\mathbb{R}} f(v)e^{-j\omega v} dv \right) \\ &= \hat{h}(\omega) \hat{f}(\omega). \end{aligned}$$

□

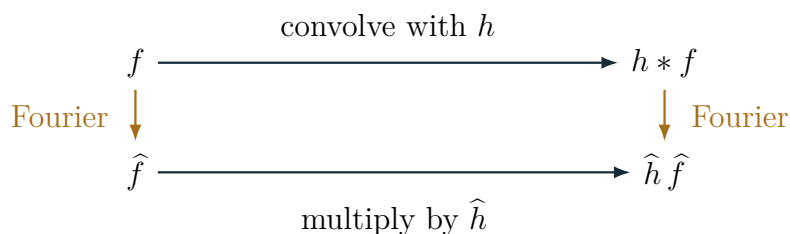


Figure 2.9: The time-domain and frequency-domain descriptions give the same output. The absolute-integrability assumptions above are sufficient conditions, not a claim that they are necessary for every useful filter.

2.10 Fourier, Laplace, and the complex plane

The *two-sided Laplace transform* uses a complex parameter s :

$$F(s) := \int_{\mathbb{R}} f(t)e^{-st} dt. \quad (2.16)$$

Write $s = \sigma + j\omega$, with $\sigma, \omega \in \mathbb{R}$. Then

$$e^{-st} = e^{-\sigma t} e^{-j\omega t}, \quad F(\sigma + j\omega) = \widehat{f(\cdot)e^{-\sigma(\cdot)}}(\omega).$$

The real part σ introduces an exponential weight; the imaginary part ω specifies oscillation. Whenever the defining integral converges on the imaginary axis,

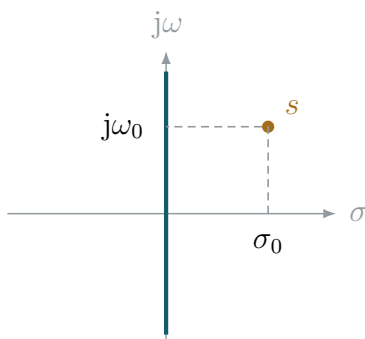
$$F(j\omega) = F(s)|_{s=j\omega} = \hat{f}(\omega). \quad (2.17)$$

The *region of convergence* is the set of s where the transform exists; for absolute convergence, require $\int_{\mathbb{R}} |f(t)|e^{-\sigma t} dt < \infty$. A Laplace transform may exist away from the imaginary axis without an ordinary Fourier integral existing on that axis.

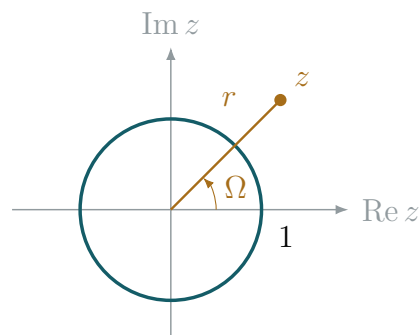
For comparison, the z -transform uses

$$X(z) = \sum_{n \in \mathbb{Z}} x[n]z^{-n}, \quad z = re^{j\Omega}, \quad r > 0,$$

so $z^{-n} = r^{-n}e^{-j\Omega n}$. The DTFT is the restriction to $r = 1$, when the series converges there. We use Ω in this comparison for angular frequency per sample.



Fourier: imaginary axis



DTFT: unit circle

Figure 2.10: Laplace separates decay and oscillation through Cartesian coordinates (σ, ω) . The z -transform separates them through polar coordinates (r, Ω) . In both cases, removing the exponential weight leaves a Fourier transform.

This geometric comparison does not identify the two transforms: one acts on functions and uses an integral; the other acts on sequences and uses a sum.

2.11 The Fourier transform of the box is a sinc

Return to $h(t) = \mathbf{1}_{[-1/2, 1/2]}(t)$. For $\omega \neq 0$, every step of the calculation is elementary:

$$\begin{aligned}\hat{h}(\omega) &= \int_{-1/2}^{1/2} e^{-j\omega t} dt \\ &= \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-1/2}^{1/2} \\ &= \frac{e^{-j\omega/2} - e^{j\omega/2}}{-j\omega} \\ &= \frac{-2j \sin(\omega/2)}{-j\omega} = \frac{\sin(\omega/2)}{\omega/2}.\end{aligned}\tag{2.18}$$

At $\omega = 0$, compute the integral directly: $\hat{h}(0) = \int_{-1/2}^{1/2} 1 dt = 1$. With the normalized convention

$$\text{sinc}(v) := \begin{cases} \frac{\sin(\pi v)}{\pi v}, & v \neq 0, \\ 1, & v = 0, \end{cases}$$

we obtain $\hat{h}(\omega) = \text{sinc}(\omega/(2\pi))$, not $\text{sinc}(\omega)$.

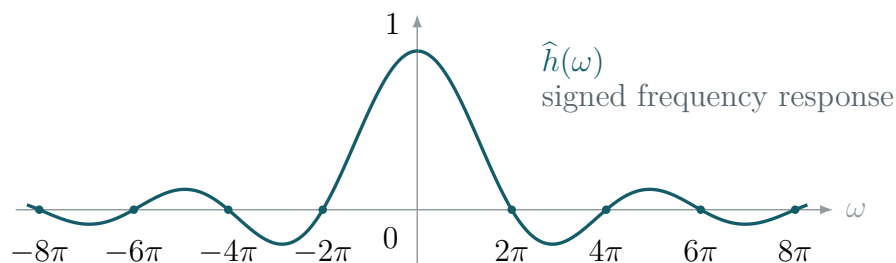


Figure 2.11: The zeros are $\omega = 2\pi k$ for nonzero integers k . The negative sidelobes are negative *values* of the response; the magnitude response is $|\hat{h}(\omega)|$.

Why is this a lowpass filter? It preserves constants because $\hat{h}(0) = 1$. Also

$$|\hat{h}(\omega)| \leq 1, \quad |\hat{h}(\omega)| \leq \frac{2}{|\omega|} \quad (\omega \neq 0),$$

by the integral triangle inequality and $|\sin(\omega/2)| \leq 1$, respectively. Rapid oscillations are attenuated. This is *not* an ideal cutoff: sidelobes remain at arbitrarily high frequencies. A negative response changes a cosine's sign, equivalently shifting its phase by π .

2.12 The derivative of an impulse

An ideal differentiator is linear and time invariant, so we expect an impulse-response description. Its impulse response should be δ' . But δ is not an ordinary function: what does its derivative mean?

Let $\varepsilon > 0$ and form the difference of two shifted impulses:

$$q_\varepsilon(t) := \frac{\delta(t + \varepsilon) - \delta(t)}{\varepsilon}. \quad (2.19)$$

The positive impulse is at $-\varepsilon$, because $t + \varepsilon = 0$ there. The negative impulse is at zero. Their total weight is zero.

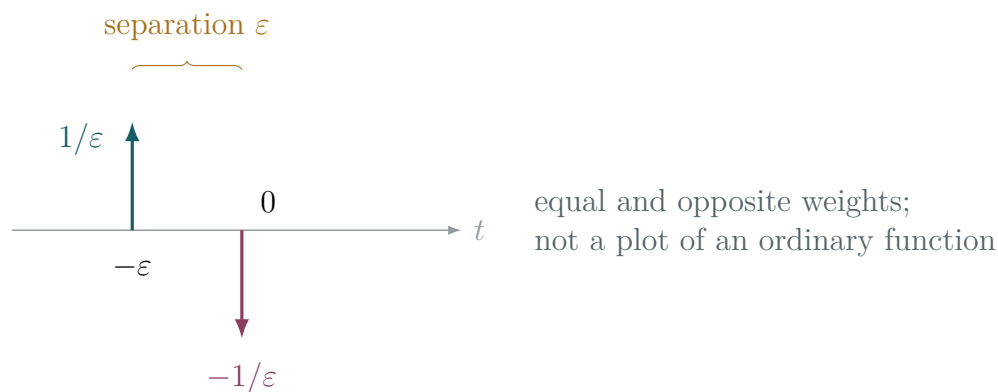


Figure 2.12: A difference quotient of impulses. As the separation shrinks, the weights increase. To understand the limit, first integrate against a differentiable function, then let the separation shrink.

Let φ be differentiable at zero. Applying sifting *before taking a limit* gives

$$\begin{aligned} \int_{\mathbb{R}} \varphi(t) q_\varepsilon(t) dt &= \frac{\varphi(-\varepsilon) - \varphi(0)}{\varepsilon} \\ &= -\frac{\varphi(-\varepsilon) - \varphi(0)}{-\varepsilon} \longrightarrow -\varphi'(0). \end{aligned} \quad (2.20)$$

We therefore define the derivative impulse by

$$\boxed{\int_{\mathbb{R}} \varphi(t) \delta'(t) dt = -\varphi'(0).} \quad (2.21)$$

Writing $q_\varepsilon \rightarrow \delta'$ means that the displayed integrals converge for every φ differentiable at zero. We evaluate each integral first, then take the limit of the resulting numbers; no pointwise limit of impulses is used. The name *doublet* refers to this limiting difference of opposite impulses.

2.13 Integration by parts and differentiation by convolution

The minus sign in (2.21) is the same minus sign that appears in integration by parts. To see the connection, first take ordinary smooth functions r and φ , with r zero outside a finite interval. Integrating the product rule gives

$$\begin{aligned}\int_{\mathbb{R}} r'(t)\varphi(t) dt &= [r(t)\varphi(t)]_{-\infty}^{\infty} - \int_{\mathbb{R}} r(t)\varphi'(t) dt \\ &= - \int_{\mathbb{R}} r(t)\varphi'(t) dt.\end{aligned}$$

The boundary term vanishes because r vanishes outside that interval. For the Dirac impulse, the corresponding derivative rule is

$$\int_{\mathbb{R}} \delta'(t)\varphi(t) dt := - \int_{\mathbb{R}} \delta(t)\varphi'(t) dt = -\varphi'(0).$$

This *definition through integration by parts* agrees with the difference-quotient limit. The result depends only on $\varphi'(0)$, so φ need not vanish outside a finite interval. There is no boundary value of δ to evaluate.

Why does convolution have no minus sign? For differentiable f , fix t and apply (2.21) to $\varphi_t(u) = f(t - u)$. The chain rule gives $\varphi'_t(u) = -f'(t - u)$. Hence

$$\begin{aligned}(\delta' * f)(t) &= \int_{\mathbb{R}} \delta'(u)f(t - u) du \\ &= - \left. \frac{d}{du} f(t - u) \right|_{u=0} = -(-f'(t)) = f'(t).\end{aligned}\tag{2.22}$$

One minus sign comes from differentiating the impulse; the other comes from the reversed argument $t - u$. They cancel.



Figure 2.13: Differentiation is convolution with the derivative impulse. Here $(Df)(t) = f'(t)$.

The same conclusion follows directly from (2.19):

$$(q_\varepsilon * f)(t) = \frac{f(t + \varepsilon) - f(t)}{\varepsilon} \longrightarrow f'(t).$$

Thus the two-impulse filter is a forward-difference differentiator before taking the limit, and δ' is its ideal continuous-time limit.

2.14 Higher derivatives and their frequency responses

For an integer $n \geq 0$, let $\delta^{(n)}$ denote the n th derivative impulse, with $\delta^{(0)} = \delta$. For a function φ with n derivatives near zero, repeated integration by parts gives

$$\int_{\mathbb{R}} \delta^{(n)}(t) \varphi(t) dt = (-1)^n \varphi^{(n)}(0). \quad (2.23)$$

Here is the induction explicitly. The formula for $n = 0$ is sifting. If it holds for n , the derivative rule gives

$$\int \delta^{(n+1)} \varphi = - \int \delta^{(n)} \varphi' = -(-1)^n (\varphi')^{(n)}(0) = (-1)^{n+1} \varphi^{(n+1)}(0).$$

Convolution. For an n -times differentiable input,

$$\begin{aligned} (\delta^{(n)} * f)(t) &= (-1)^n \left. \frac{d^n}{du^n} f(t-u) \right|_{u=0} \\ &= (-1)^n (-1)^n f^{(n)}(t) = f^{(n)}(t). \end{aligned} \quad (2.24)$$

Fourier transform. Apply the same rule to $\varphi(t) = e^{-j\omega t}$:

$$\begin{aligned} \widehat{\delta^{(n)}}(\omega) &= (-1)^n \left. \frac{d^n}{dt^n} e^{-j\omega t} \right|_{t=0} \\ &= (-1)^n (-j\omega)^n = (j\omega)^n. \end{aligned} \quad (2.25)$$

As for δ , we evaluate the Fourier integral using the impulse rule with an exponential. It is not an ordinary absolutely convergent integral.

Keep the three identities separate

$$\begin{aligned} \int_{\mathbb{R}} \delta^{(n)}(t) \varphi(t) dt &= (-1)^n \varphi^{(n)}(0), \\ (\delta^{(n)} * f)(t) &= f^{(n)}(t), \\ \widehat{\delta^{(n)}}(\omega) &= (j\omega)^n. \end{aligned}$$

The sign in the integration identity cancels in convolution through the chain rule.

For $n = 2$, the multiplier is $(j\omega)^2 = -\omega^2$, exactly as in $\frac{d^2}{dt^2} \cos(\omega t) = -\omega^2 \cos(\omega t)$. Unlike the box average, an ideal differentiator amplifies high frequencies: its gain is $|\omega|^n$ for $n \geq 1$.

Next: sampling and sinc interpolation. The box-to-sinc pair will reappear with time and frequency exchanged. We will connect reconstruction with minimum-energy interpolation and projection.